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RESEARCH INTERESTS

My research aims to understand and predict how soils store carbon and sustain ecosystem functioning in tropical landscapes under global change. I combine Earth observation, machine learning and spatio-temporal statistical modelling to advance digital soil mapping (DSM) and the spatial prediction of soil properties and carbon stocks, from open ecosystems and agroforestry systems to blue carbon in mangroves. At finer scales, I study soil carbon and nitrogen biogeochemistry, soil organic matter dynamics and organo-mineral interactions, as well as the links between soil health, soil ecology and functional ecology in the Anthropocene. I am particularly interested in how land management shapes soil carbon, and in soil–plant–atmosphere feedbacks in tropical and semi-arid agroecosystems. Through precision agriculture and UAV-based remote sensing, I also seek to turn these approaches into robust evidence for sustainable land management and climate policy.

EDUCATION

Double Master's Degree Program *2023–2026*
Jointly enrolled in two Master's programs in parallel between the University of Antananarivo and Sorbonne Université.

Master of Science in Biodiversity, Ecology, and Evolution (BEE) *2024–2026*
Sorbonne Université, MNHN, Paris-Saclay University, Paris, France
Specialization: Evolutionary and Functional Ecology

Master of Science in Tropical Agriculture and Sustainable Development (AT2D) *2023–2025*
University of Antananarivo, School of Agronomic Sciences (ESSA), Madagascar
Specialization: Soil Biofunctioning and Environment

Bachelor of Science in Agronomic and Environmental Sciences *2020–2023*
University of Antananarivo, ESSA, Madagascar

RESEARCH EXPERIENCE

Research Collaborator – Taneti Project (Soil Carbon in Open Ecosystems) *2026–Present*
Royal Botanic Gardens, Kew (UK) & Laboratory of Radio-Isotopes, University of Antananarivo, Madagascar

- Compiled and harmonized a multi-decadal national soil database for Madagascar to update national soil organic carbon (SOC) data, prioritizing subsoil data first and then the full 0–100 cm profile.
- Applied machine learning and spatial modelling with Earth observation covariates to predict and map SOC stocks and to update the national soil carbon map.
- Developed a soil fertility indicator (SFI), a composite index derived from soil physico-chemical properties in open ecosystems, and modelled and mapped it using machine learning.
- Analysed the environmental controls on fire effects on SOC across forest–grassland mosaics.
- Contributed to four manuscripts arising from this work (see Publications).

Collaborator – FAO & The Nature Conservancy (TNC) *2026–Present*
Food and Agriculture Organization of the United Nations (FAO) & The Nature Conservancy (TNC)
Leads: Martial Bernoux (FAO) and Susan Cook-Patton (TNC)
In collaboration with R. Cardinael

- Quantifying the effects of the IPCC input (F_I) and management (F_{MG}) stock change factors on SOC in agroforestry systems, with a primary focus on F_I, to estimate the potential for carbon restitution to the ecosystem when cropping systems are converted to agroforestry.
- Assessing the effect of management practices (fertilizer, tillage, irrigation, litter) on SOC using the TNC SOC Agroforestry database.
- Performed SOC depth harmonization and calculated F_I and F_{MG}.
- One paper in preparation.

Research Assistant – Mangrove Carbon Stocks Modelling *Apr. 2026–Present*
Laboratory of Radio-Isotopes, University of Antananarivo, Madagascar
Lead: Herintsitohaina Razakamanarivo

- Modelling and predicting SOC, aboveground biomass (AGB) and belowground biomass (BGB) in Madagascar's mangroves at national scale, integrating machine learning and remote sensing.

- Collaborating with blue carbon and mangrove stakeholders in Madagascar to collect field data, and harmonizing these heterogeneous datasets for modelling through dedicated quality-control analyses and allometric/predictive equations.
- Quantifying carbon stocks, identifying key environmental drivers and spatio-temporal dynamics, and supporting international climate frameworks including REDD+.

External Research Collaborator – Smart Agriculture, Cotton Crop Systems *Dec. 2024–Present*
Institute of Cotton Research, Chinese Academy of Agricultural Sciences, China
Lead: Yabing Li

- Contributed to doctoral research on precision agriculture in cotton systems, designing UAV-based remote sensing methods to improve resource-use efficiency and crop management.
- Developed and benchmarked object-based classification models using UAV multispectral and RGB imagery for crop-type mapping across phenological stages.
- Developed and benchmarked machine learning regression models for stage-specific cotton yield estimation.
- Contributed to three papers (see Publications).

Research Intern – Opti-Ferti-CSS Project *Aug. 2025–Jan. 2026*
CIRAD, Recycling and Risk Research Unit & Aïda Research Unit, Montpellier, France
Leads: Matthieu Bravin and Mathias Christina

- Investigated biophysical and management determinants of sugarcane productivity in Sahelian irrigated agroecosystems to guide targeted yield-enhancing interventions.
- Integrated 21 years of climate, soil and crop management data in a statistical and machine learning framework to predict aboveground biomass, sucrose content and recoverable sugar per hectare.
- Applied Integrated Nested Laplace Approximation (INLA) for Bayesian spatio-temporal interpolation of soil properties, filling spatial sampling gaps and improving predictive accuracy.
- One paper in preparation.

Research Intern – SoilPACMAN Project *Mar.–May 2025*
iEES Paris (Institute of Ecology and Environmental Sciences of Paris), France
Leads: Naoise Nunan, Xavier Raynaud and Steffen A. Schweizer (ANR & DFG funded project)
Soil Pores affecting Carbon Mineralisation at nanoscales: how soil organic carbon dynamics are locally controlled by the organo-mineral microenvironment of microbes in the pore space

- Investigated how pore-scale soil architecture governs organic carbon accessibility and microbial carbon processing in a luvisol.
- Used X-ray microtomography (UCT) image slices to simulate the spatial distribution and accessibility of particulate organic matter (POM) and dissolved organic carbon (DOC), as well as the distribution of microbial necromass, across micropore and macropore networks.
- Modelled differential substrate exposure for copiotrophic vs. oligotrophic microbial guilds to assess carbon use efficiency (CUE) as a function of pore microenvironment, linking soil physical architecture to carbon biogeochemical outcomes.
- Contributed to one paper and two presentations.

PUBLICATIONS

Peer-Reviewed Articles

Be, M. C., Zhang, J., Yang, B., **Randrianantenaina, A. S.**, et al. (2026). Using UAV-based RGB and multispectral imagery to estimate cotton above-ground biomass by integrating multi-modal features and machine learning algorithms. *Remote Sensing*, *18*(14), 2278. https://doi.org/10.3390/rs18142278

Be, M. C., Yang, B., **Randrianantenaina, A. S.**, et al. (2026). Comparative analysis of deep learning architecture for object-based crop classification using UAV multispectral imagery. *Remote Sensing Applications: Society and Environment*, *44*, 102295. https://doi.org/10.1016/j.rsase.2026.102295

Be, M. C., **Randrianantenaina, A. S.**, Kanneh, J. E., et al. (2025). Comparative analysis of machine learning algorithms for object-based crop classification using multispectral imagery. *Drones*, *9*, 763. https://doi.org/10.3390/drones9110763

Maestrali, M., Nunan, N., **Randrianantenaina, A. S.**, et al. (2026). Microbial carbon use efficiency emerges from interactions between soil structure and life-history strategies. *SOIL*, *12*(2), 1079–1099. https://doi.org/10.5194/soil-12-1079-2026

Submitted & Under Review

Randrianantenaina, A. S., Ramifehiarivo, N., Andriamananjara, A., et al. (2026). First national high-resolution mapping of Madagascar's mangrove carbon stocks using Earth observation. *Submitted to International Journal of Applied Earth Observation and Geoinformation*.

Ramifehiarivo, N., **Randrianantenaina, A. S.**, Ratovoarimanana, L. M., et al. (2026). Update of the national soil carbon map of Madagascar using additional data and earth observation platforms. *Under review in Soil Advances*.

Ratovoarimanana, L., Andriamananjara, A., Ramifehiarivo, N., et al., **Randrianantenaina, A. S.**, et al. (2026). Environmental factors regulating fire effects on soil organic carbon stocks across forest–grassland mosaics in Madagascar. *Submitted to Fire Ecology*.

Ramifehiarivo, N., Ratovoarimanana, L. M., Andriamananjara, A., **Randrianantenaina, A. S.**, et al. (2026). Towards a better understanding of spatial carbon storage in open ecosystem areas. *Under review in Geoderma Regional*.

In Preparation

Randrianantenaina, A. S., Christina, M., Heuclin, B., et al. (2026). Assessing quantitative vs. qualitative trade-offs in sugarcane yield: a 20-year, farm-scale study in the Senegalese Sahel using machine learning. *Targeted journal: Field Crops Research*.

Rafidimanantsoa, L. S., Ramifehiarivo, N., Ratovoarimanana, L., et al., **Randrianantenaina, A. S.**, et al. (2026). Spatial prediction of soil fertility across tropical forest–grassland mosaics. *Targeted journal: Geoderma Regional*.

Preprints

Maestrali, M., Nunan, N., **Randrianantenaina, A. S.**, et al. (2026). Microbial carbon use efficiency emerges from interactions between soil structure and life-history strategies. *EGUsphere*. https://doi.org/10.5194/egusphere-2026-1818

Datasets

Randrianantenaina, A. S. (2026). *Mangrove forest distribution and dynamics in Madagascar (1995–2025)* (Version v1) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.18620206

SKILLS

Digital Soil Mapping & Soil Science: Machine learning-based DSM of soil organic carbon and soil properties using Earth observation covariates; model validation and uncertainty assessment; soil health and soil fertility assessment; soil database compilation and depth harmonization; soil carbon stock change factors (F_I, F_{MG})

Modelling & Statistics: Machine learning and deep learning; Bayesian spatio-temporal modelling (INLA); geostatistics; ecological and crop modelling; stochastic processes; high-performance computing (HPC)

Remote Sensing & GIS: UAV multispectral and RGB image analysis; object-based image classification; satellite remote sensing; Google Earth Engine; QGIS; ArcGIS Pro

Programming & Data Management: Python, R, JavaScript; parallel computing; workflow automation; multi-source data harmonization and quality control

Field & Agronomy: Soil characterization; field experimentation; irrigation and fertilization management; agronomic diagnostics

Laboratory: Soil physico-chemical analysis (pH, texture, CEC, mineral nitrogen: NO_2^-, NO_3^-, NH_4^+); ^{13}C isotopic labelling of soil organic matter; soil respiration quantification

Stakeholder Engagement & Collaboration: Building and sustaining partnerships with national and international stakeholders (research institutes, NGOs, local and conservation actors); coordinating multi-source data collection; working effectively in interdisciplinary, multicultural and multilingual teams

Communication & Leadership: Scientific writing; presenting results to diverse audiences; technical workshops (R, Python, GIS); technical translation; scientific outreach; promotion of sustainable agriculture

Professional Competencies: Project and time management; problem solving; critical thinking; adaptability; autonomy and initiative; attention to detail

Languages: Malagasy (native); French (professional); English (professional)

Certifications & Networks: Soil Health Training (Cornell University); Ecological Modeling; EO Africa Community member

REFERENCES

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